Roll No.....

End Semester Examination 2025

Name of the Course: M.Tech (VLSI Design and Systems) **Semester: IIIrd**
Name of the Paper: Organic Electronic Devices and Circuits **Course Code: VDM 341**
Time: 03 Hr. **Maximum Marks: 100**

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.

Each question carries 10 marks

Q1	(20 marks)	**C01,C02, CO6**
(a)	Explain the differences between organic and inorganic semiconductors. Highlight the advantages and disadvantages of organic semiconductors, as well as the restrictions they face when compared to inorganic semiconductor	
(b)	Describe the organic materials-based humidity and pH sensors with applications.	
(c)	Discuss the energy band diagram in organic semiconductors and its relevance to the charge transport process. How does charge transport occur in organic semiconductors?	
Q2	(20 marks)	**C02,C03,C06**
(a)	What are the merits and demerits of OLED technology? How have OLEDs found applications in the field of display technology? Provide examples of consumer electronics and lighting solutions where OLEDs are commonly used.	
(b)	Why are stability issues a concern for OLED technology, and what factors contribute to the degradation of OLED devices over time.	
(c)	Compare the performance of Single Gate (SG) and Dual Gate (DG) Thin Film Transistors. What are the advantages and limitations of each configuration?	
Q3	(20 marks)	**C02 ,C05**
(a)	Explain the concept of organic complementary inverter circuits and their role in electronic devices. How do they differ from traditional inorganic inverters?	
(b)	Define organic piezoelectric pressure sensors .Also Describe the industrial applications.	
(c)	Define organic piezoresistive pressure sensors and elaborate on their industrial applications.	
Q4	(20 marks)	**CO3,C05,C06**
(a)	Explain the organic material-based capacitive sensors with applications.	
(b)	What is the fundamental principle behind the operation of organic solar cells, and how do they convert sunlight into electrical energy? Also Describe the key characteristics of organic solar cells	
(c)	Describe the organic material-based temperature sensors with applications.	
Q5	(20 marks)	**C04,CO5**
(a)	Compare organic and conventional sensor, mentioning their benefits, drawbacks, and limitations.	
(b)	Briefly explain the basics of ionic polymer-metal composites (IPMC) and their potential applications in sensing technology.	
(c)	What are the different classification of organic solar cells? Also mention different applications .	